

**BEFORE YOU READ — ANSWER FROM MEMORY**

1. Which two GI enzymes are secreted as inactive zymogens, and what activates each?
2. Which GI hormone was the first ever discovered, and what does it do to pancreatic bicarbonate secretion?
3. Why does undigested lactose cause diarrhoea and bloating?

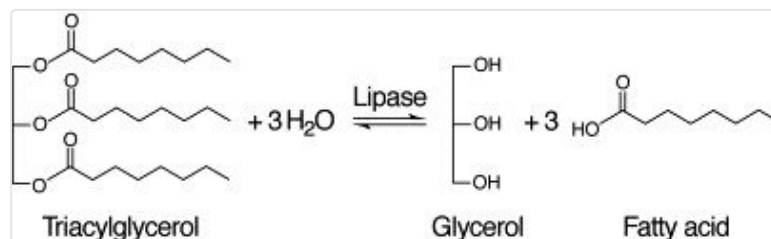
Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Enzymes — general properties

What they are	Water-soluble, highly specialised proteins — biocatalysts that speed up biochemical reactions without being consumed
How	Lower the activation energy of the reaction; don't change the nature/properties of the end product
Specificity	Highly specific — each enzyme typically catalyses one kind of substrate
Ribozymes	Special RNA species that also act as enzymes; catalysis via transesterification & phosphodiester bond hydrolysis; inactivated by heating past melting temperature or by changing essential nucleotides

2 Digestive enzymes — overview

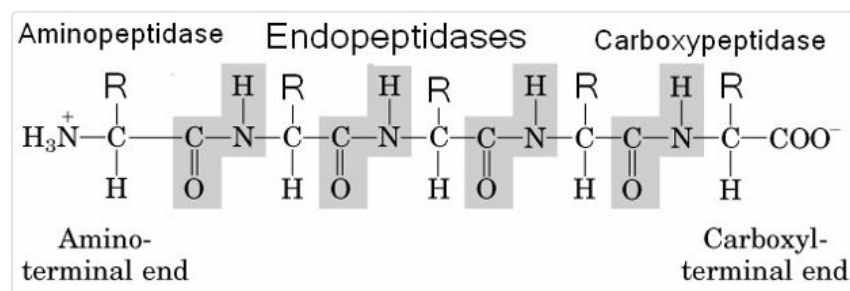
Enzyme	Substrate → product	Source
Amylase	Starch → dextrins, maltose, glucose	Mainly salivary gland & pancreas; weak activity in liver, lung, fallopian tube, ovary, prostate
Lipase	Fats → fatty acids + glycerol	Small amounts from mouth & stomach; large amounts from pancreas
Proteases	Proteins → polypeptides / free amino acids	Stomach & pancreas



Lipase hydrolyses triacylglycerol + water into glycerol + 3 fatty acids.

3 Proteases — the core discriminator

Endopeptidases cut **inner** peptide bonds; exopeptidases trim from an **end** — aminopeptidases from the N-terminus, carboxypeptidases from the C-terminus.



Aminopeptidases act at the N-terminal end; carboxypeptidases at the C-terminal end; endopeptidases cut internal bonds.

Enzyme	Type	Cleaves at / specificity	Source & notes
Pepsin	Endopeptidase	Peptide bonds of aromatic or acidic amino acids	Stomach; secreted as inactive pepsinogen , activated by HCl or autocatalytically by active pepsin

Rennin	—	Milk protein (casein)	Stomach; digestion of milk in infants
Trypsin	Endopeptidase	Where the carbonyl group is from Arg or Lys	Pancreas; also activates other pancreatic enzymes (carboxypeptidase, chymotrypsin)
Chymotrypsin	Endopeptidase	Trp, Tyr, Phe, Met, Leu	Pancreas
Carboxypeptidase A	Exopeptidase (C-terminal)	Ala, Ile, Leu, Val	Pancreas
Carboxypeptidase B	Exopeptidase (C-terminal)	Arg, Lys (basic AAs)	Pancreas
Elastase	Endopeptidase	Ala, Gly, Ser	Pancreas

4 Enzyme deficiency — causes & named conditions

Common causes	Diet high in sugar/processed foods, repeated antibiotic use, smoking, aging, pregnancy, physical/psychological/emotional stress		
Signs	Food intolerances, abdominal pain, diarrhoea & bloating, constipation, undigested food in stool, poor nutrient absorption, weaker immunity, fatigue/anxiety/depression, skin problems		
Condition	Mechanism	Management / associations	
Lactase deficiency	Undigested lactose ↑ osmotic pressure (draws water into the lumen → diarrhoea) + bacterial fermentation of lactose → CO ₂ gas → abdominal cramps	Remove lactose (milk) from the diet	
Exocrine pancreatic insufficiency (EPI)	Deficiency of amylase, protease & lipase from the pancreas → cannot digest food properly, especially fats	Associated with chronic pancreatitis, cystic fibrosis, pancreatic cancer	

5 Hormones — general properties

What they are	Chemical messengers released by a cell/gland, sent to affect cells elsewhere in the body
General features	Secreted in very small amounts; usually transported in blood; target cells have specific receptors; regulate cell reactions by affecting gene expression; regulate metabolic events
GIT hormones	Belong to the "local hormone" family; control GI secretion & regulate GIT motility; mostly peptides; many also found in other tissues, especially the brain

6 The GIT hormones — the core discriminator

Hormone	Source	Stimulus for release	Main actions
1. Gastrin	γ cells — stomach, duodenum, pancreas	Peptides & amino acids in the gastric lumen	↑ gastric juice (HCl + pepsin) secretion; ↑ gastric motility; proliferation of gastric epithelium
2. Cholecystokinin (CCK)	Duodenum & jejunum	Fat digestion products, fatty acids, protein, monoglycerides	Contracts gallbladder → bile release; inhibits stomach contraction & slows gastric emptying; stimulates pancreatic enzyme release
3. Secretin	Duodenum & upper jejunum	Acidic gastric juice emptying into the duodenum	↑ pepsin secretion; ↓ gastric acid (inhibits gastrin); stimulates bile secretion & gallbladder contraction; stimulates pancreatic bicarbonate secretion; enhances CCK's effect; ↑ insulin release after oral glucose; regulates water homeostasis
4. Somatostatin	δ cells — stomach, upper small intestine, pancreatic islets	Chyme with glucose & protein in stomach/ small intestine	Inhibits GH & TSH (anterior pituitary); inhibits gastric HCl secretion & motility; inhibits pancreatic juice secretion; inhibits insulin & glucagon; inhibits gastrin, CCK, motilin
5. GIP	Duodenum & jejunum mucosa	—	Inhibits gastric acid secretion & gastric motility

6. Ghrelin	Endocrine cells of the stomach	Hunger	Acts on the hypothalamus to stimulate feeding; counteracts leptin's inhibition of feeding
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Secretin was the **first GI hormone discovered**. Note the division of labour: secretin drives pancreatic **bicarbonate**; CCK drives pancreatic **enzymes** and gallbladder contraction — and each enhances the other's effect.

P Must-remember pearls

- Pepsin (stomach) and trypsin (pancreas) are both secreted as inactive zymogens — pepsinogen and trypsinogen.
- Trypsin cleaves after Arg/Lys; chymotrypsin after Trp/Tyr/Phe/Met/Leu; carboxypeptidase A releases neutral/branched AAs; carboxypeptidase B releases Arg/Lys, from the C-terminus.
- **Secretin** = first GI hormone discovered; drives pancreatic **bicarbonate**. **CCK** drives pancreatic **enzymes** + gallbladder contraction. They enhance each other.
- **Somatostatin** is the master inhibitor: turns down GH, TSH, gastric acid/motility, pancreatic juice, insulin, glucagon, and other GI hormones (gastrin, CCK, motilin).
- Lactase deficiency → undigested lactose pulls water osmotically (diarrhoea) + bacterial fermentation (gas/cramps); managed by removing lactose from the diet.

Q Test yourself

1. Which enzyme is secreted as an inactive zymogen and activated by HCl or autocatalytically by other active enzyme molecules?

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|-------------|------------|
| A) Trypsin | B) Pepsin |
| C) Elastase | D) Amylase |

2. Trypsin cleaves peptide bonds where the carbonyl group is contributed by:

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|-------------------------|---------------|
| A) Aromatic amino acids | B) Arg or Lys |
| C) Ala or Gly | D) Trp or Tyr |

3. Which hormone was the first GI hormone discovered, and stimulates pancreatic bicarbonate secretion?

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|-------------|-----------------|
| A) Gastrin | B) CCK |
| C) Secretin | D) Somatostatin |

4. A hormone that inhibits insulin, glucagon, gastric acid secretion, and several other GI hormones is:

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|------------|-----------------|
| A) Gastrin | B) GIP |
| C) Ghrelin | D) Somatostatin |

5. Undigested lactose in the gut causes diarrhoea mainly because it:

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| A) Directly damages the intestinal epithelium | B) Increases osmotic pressure, drawing water into the lumen |
| C) Blocks bile secretion | D) Inhibits pancreatic amylase |

ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1. B — Pepsin** — secreted as pepsinogen; activated by HCl or autocatalytically.
- 2. B — Arg or Lys** — trypsin's defining cleavage specificity.
- 3. C — Secretin** — first GI hormone discovered; drives pancreatic bicarbonate secretion.
- 4. D — Somatostatin** — the broad inhibitor of GI/endocrine secretion.
- 5. B — Increases osmotic pressure** — unabsorbed lactose draws water into the bowel lumen; bacterial fermentation adds gas/cramps.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Biochemistry GIT Block · MEDucation by Mattin Majid · GIT Enzymes and Hormones, Dr Sardar N. Ahmed
Study alongside the lecture, not instead of it.